

ECON 1020: Principles of Microeconomics

Homework 1: Chapters 1–2

Ten Principles of Economics & Thinking Like an Economist

Due on Monday, September 14 at 12:20 PM

Answer every question. This homework is graded on completion, not correctness, but you should still do your best: your TA will go over the answers, and this material is fair game for quizzes and exams.

Submissions are accepted either online through NTU COOL or as a hard copy at your recitation session.

Part A: Ten Principles of Economics

1. One trade-off society faces is between efficiency and equality. Define each term. If the government redistributes income from the rich to the poor, explain how this action affects equality as well as efficiency in the economy.

2. You have the afternoon free. You have a choice between going to the movies with a friend or studying economics for three hours. If you go to the movies, you will spend NT\$350 on a ticket and NT\$150 on popcorn. If you choose to study economics for three hours, you will raise your exam grade by 15 points.
 - (a) What is your opportunity cost of going to the movies?

 - (b) What is your opportunity cost of studying economics?

3. Wei-Ting quits her job at a tech firm, where she earns NT\$1,200,000 a year, to pursue a three-year doctoral program. Her annual school expenses are NT\$120,000 for tuition, NT\$20,000 for books, NT\$180,000 for rent, and NT\$100,000 for food. What is her opportunity cost of attending the three-year doctoral program?

4. It costs a company NT\$1,440,000 to produce 1,200 smartwatches. If the company produces one more smartwatch, the total cost rises to NT\$1,441,350. The company is currently producing 1,200 smartwatches.
- (a) What is the company's average cost per smartwatch?
 - (b) What is the company's marginal cost of the 1,201st smartwatch?
 - (c) A customer is willing to pay NT\$1,650 for the 1,201st smartwatch. Should the company produce and sell it? Explain using the concepts of marginal cost and marginal benefit.
5. People respond to incentives. Suppose a school extends the school year from 9 months to 11 months, with no change in teachers' total pay. Using the concepts of marginal cost and marginal benefit, explain how this change might affect teachers' incentive to stay in the profession.

Part B: Thinking Like an Economist

6. For each topic below, say whether it belongs to microeconomics or macroeconomics.
- (a) the impact of a change in consumer income on the purchase of luxury automobiles
 - (b) the impact of a war in the Middle East on Taiwan's inflation rate
 - (c) factors influencing Taiwan's rate of economic growth
 - (d) the effect of pollution taxes on Taiwan's semiconductor industry

(e) the level of competition among streaming service providers

7. For each statement below, say whether it is positive or normative, and briefly explain how you can tell.

(a) The minimum wage creates unemployment among young and unskilled workers.

(b) The minimum wage ought to be abolished.

(c) If the price of a product in a market decreases, then, other things equal, quantity demanded will increase.

(d) Taiwan's income distribution is not fair.

(e) There is a trade-off between inflation and unemployment in the short run.

8. An economy produces only two goods, smartphones and pizzas. The table below shows some of the combinations it can produce with its current resources and technology.

Smartphones	Pizzas
0	10,000
600	8,000
1,100	6,000
1,500	4,000
1,800	2,000
2,000	0

(a) What is the opportunity cost of increasing smartphone production from 600 to 1,100?

(b) What is the opportunity cost of increasing pizza production from 4,000 to 6,000?

- (c) Is this production possibilities frontier a straight line or bowed outward? How can you tell from the table?

9. Answer each of the following about the circular-flow diagram.

- (a) Who owns the factors of production and consumes all of the goods and services produced in the economy?
- (b) When a worker provides labor to a firm through the market for factors of production, what is the flow of money he receives in exchange called?
- (c) When a household buys a good in the market for goods and services, what does that household's spending become for the firm that sold it?
- (d) Name one element of the real economy that the simple circular-flow diagram leaves out.